

Collaborative relational time without a global clock: Reconstruction, partial gluing, and gauge-invariant calibration structure in finite event networks

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Abstract. We develop a finite relational theory of temporal organization without assuming a primitive global scalar clock. Building on earlier serialization-gauge results, we first distinguish regional informational sufficiency from trajectory-dependent causal domains. Conditional on a supplied collective ordering/calibration structure, finite validation shows that a global scalar coordinate and total serialization need not be primitive; separate local-clock and calibration constructions then reconstruct collective chronology across exact, dynamically changing, mixed-record, and approximate-closure layers. We next define collaborative complexity as the minimum number of local views needed to determine a target temporal relation. Local token/ORD microdynamics generate nontrivial collaborative complexity without a global target oracle, and a one-shot 288-case synthetic holdout supports finite robustness under sparse networks, shortcuts, connected token loss, and richer routing rules. Historical precedence is then separated from mutable active coordination; ORD and affine-CAL data obey distinct gluing criteria, and finite tests separate local relational coherence from conditioning of a chosen global affine reduction. In the positive CAL scale sector, relative calibration defines a gauge-covariant one-cochain with gauge-invariant curl and exact/coexact/harmonic decomposition. Projection away from exact directions yields gauge-invariant nonintegrable content for a fixed cochain inner product. Given supplied provenance and a declared density convention, one obtains a gauge-invariant current assignment and exact discrete balance law. The results are finite and structural and do not establish universal temporal emergence, continuum spacetime, Lorentz symmetry, gravity, physical curvature, or a Noether charge.

I. INTRODUCTION

The predecessor article asked when different serializations of a finite event process are gauge-equivalent and when conditioned hazards obstruct that equivalence, as established in Ref. 13. The present article takes the next question rather than repeating the first one: after serialization has ceased to be the central representation problem, what temporal organization can be built from local event relations themselves, and which parts of that organization require genuinely joint information? We work entirely in finite relational models and do not assume a primitive global scalar clock.

The separation between event ordering and clock representation has antecedents in distributed event ordering (Ref. 1), while relational and clock-free formulations motivate complementary questions about temporal description (Refs. 2 and 3). Finite event partial orders are also mathematically adjacent to causal-set constructions (Ref. 14), but no causal-set continuum approximation or spacetime interpretation is assumed here. The present construction is narrower: it concerns finite relational event systems, explicit reconstructibility, and falsification-oriented finite validation rather than a universal ontology of time.

This continuation is deliberately nonretroactive. The predecessor separated event-poset structure, compatible serializations, conditional event kernels, observable quotients, and conditioned hazards; it did not prove that physical time is globally redundant. The present article treats those earlier results as fixed inputs and does not use later constructions to revise earlier claims, failed paths, thresholds, or equivalence notions.

The first successor question is regional and informational rather than collaborative. Operator support, regional sufficiency for a declared channel or target, accessibility of that regional information to an arbitration rule, and a trajectory-dependent causal domain are distinct notions. The regional analysis establishes informational sufficiency/obstruction over the declared admissible configuration family; it does

not claim that the witness configurations are dynamically reachable from one common origin, nor does it identify an informationally sufficient region with a full causal domain.

The second question separates representation from construction. Conditional on an already supplied collective temporal structure, a finite validation shows that a scalar global coordinate and total serialization can be removed as primitive representational inputs within the declared domain. This statement is intentionally conditional: it does not derive the collective structure from local clocks. A subsequent staged reconstruction program addresses that missing step by building chronology from local clocks and calibrations through exact factorized, dynamically changing, correlated-domain, mixed-record, and controlled approximate-closure layers.

Only after those two steps do we ask the stronger collaborative question. For a declared family of finite histories, a target temporal relation can require several local participant views for reconstruction even though no individual view determines it. Bounded local token/ORD microdynamics can generate such history families without reading a global target variable or consulting a scalar clock. In this precise sense, overlapping dependencies can support nontrivial collaborative temporal information. The phrase “collaborative time” in this paper refers to that operational property; it is not an assertion that physical time itself has been ontologically derived.

The remaining theory is built downstream of this collaborative layer. Historical precedence is separated from mutable active coordination; active coordination patches can split and re-form without historical memory acting as a hidden synchronization channel; local charts form temporal atlases whose ORD and CAL data obey different gluing criteria; finite-scale tests separate local relational coherence from the conditioning of a single scalar coordinate; and matched-connectivity controls show that ORD, CAL, and CORR relations cannot be reduced to bare graph connectivity. Only after those steps do temporal domains become a useful interaction-level concept rather than the starting ontology.

A further layer concerns integrability and gauge. Global affine CAL scalarization is treated as an optional reduction, distinct from an order-preserving serialization of an ORD partial order. When affine calibrations integrate, one common affine representation is available; when they do not, partial precedence, local calibration, holonomy, harmonic periods, and provenance-conditioned balance may still be well defined. In the positive CAL scale sector this leads to a gauge-covariant edge cochain, an exact/coexact/harmonic decomposition, a no-go for gauge-dependent raw divergence, and a provenance-conditioned finite balance law for gauge-invariant nonintegrable calibration content.

Exact finite statements and computational validation studies are kept distinct throughout. Predeclared synthetic holdouts test robustness of declared constructions, while semantic consistency checks delimit interpretation; neither is treated as experimental evidence about nature. Sec. II separates regional informational locality, conditional global-coordinate redundancy, reconstruction from local clocks/calibrations, and typed interfaces; Sec. III establishes collaborative reconstructibility and its external robustness test; Secs. IV–VII develop dynamic coordination, atlases, scalar deconditioning, and typed partial gluing; Secs. VIII–XI develop the gauge/Hodge and provenance-balance structure; Sec. XII summarizes evidential status and reproducibility; and Secs. XIII–XIV give interpretation and conclusions. Machine-oriented provenance and negative-path records are retained in Appendix B and in the reproducibility archive rather than used to drive the scientific narrative.

II. FROM REGIONAL INFORMATION TO COLLECTIVE TEMPORAL STRUCTURE

A. Relation to the predecessor and nonretroactivity

The predecessor research line is not reinterpreted here. Successor constructions may inherit its fixed mathematical objects, validated finite results, and stated limitations, but they may not retrospectively revise predecessor claims, failed paths, thresholds, or equivalence semantics. This dependency rule is part of the reproducibility protocol rather than a new temporal theorem.

This distinction is substantive. The predecessor studies serialization equivalence and obstruction for supplied event dynamics; the present article asks what finite collective temporal structure can be reconstructed and generated when local event relations and declared local views are operative inputs. Removing a primitive global scalar clock from the successor ontology is therefore a modeling choice tested downstream, not a retroactive claim about the physical content of the predecessor models.

B. Regional informational locality and the reachability boundary

The regional layer distinguishes four questions that are easily conflated: where an operation is supported, which region is informationally sufficient for a declared channel or target, whether that information is accessible to the rule that must arbitrate an update, and which events belong to a trajectory-dependent causal domain. The first three can be posed on a declared finite configuration family without already solving the fourth.

Accordingly, the regional obstruction used here is an informational statement on admissible configurations. It does not assert that two witness configurations are reachable from a common dynamical origin. “Regional locality” below therefore means channel-relative informational sufficiency/accessibility, not a proof of a trajectory-dependent causal domain. This boundary prevents an information-theoretic statement from being silently promoted to a dynamical-causality claim.

C. Conditional redundancy of a global scalar coordinate

A separate finite validation asks a conditional representation question. Suppose the collective temporal structure needed for arbitration—local temporal data together with the declared ordering/compatibility/calibration relations—is already supplied. Within the declared finite domain, the global scalar coordinate and total serialization are then not required as primitive state variables: the relevant scheduler outputs can be reconstructed from the supplied collective structure. The validation comprised 32 cases, split evenly between training and holdout, and satisfied all 16 predeclared criteria with exact replay.

The condition is essential. This result does not explain where the collective structure comes from, and it is not a proof that global physical time is absent. Its role is narrower: once the relational structure is given, one global scalar coordinate is a representation rather than an additional constitutive object. The reconstruction problem addressed next is therefore logically prior to any claim of collaborative generation.

D. Reconstruction from local clocks and calibrations

The next stage removes that supplied-structure shortcut. Reconstruction was developed incrementally so that exact factorized cases, dynamic creation and invalidation of certified interfaces, correlated-domain clocks, mixed instrument records, rollback/ownership semantics, and approximate closure were tested separately rather than folded into one emergence claim. The construction uses local clock and calibration data plus explicitly declared interfaces; it does not assume an independently available global scalar time variable.

Finite validation suites covered 60/60 exact factorized cases, 80/80 dynamic-interface cases, and 112/112 correlated-domain-clock cases; the approximate-closure/effective-macrotime layer later passed 240/240 declared cases. These stages test different objects and should not be read as one universal theorem. In particular, the interfaces and contact substrate are supplied by the model: the program does not derive autonomous discovery of arbitrary remote interactions, a continuum, or a universal many-body clock.

This staged reconstruction is the bridge between “a global coordinate is dispensable once collective structure is known” and the stronger question “can collective temporal information itself arise from bounded local dependencies?” The latter is the collaborative problem of Sec. III.

E. Typed interface composition

A finite typed interface-composition study then checked executable composition of the adapters connecting the local-clock, dynamic-domain, mixed-record, and approximate-closure layers. All 156 declared cases satisfied the composition criteria; the two independent evaluation paths agreed in 156/156 cases, and all 48

multipath concordance checks passed. Its role is implementation consistency across interfaces used later, not a theorem of time.

An earlier interface-composition attempt was invalidated by design and oracle conflicts before it could support a scientific claim. That negative path is retained in Appendix B; none of its synthetic mutations is used as a physical counterexample or as evidence for the successor theory.

F. Operational primitives and relation typing

The successor theory uses a deliberately sparse operational ontology. Events and local precedence are primitive inputs. Local read/write or provenance information is admitted only when explicitly supplied by the micro-event law, and global closure, connected components, scheduler variables, aggregate statistics, and adjudication objects remain analyst-level objects rather than microdynamic oracles.

Optional local clock coordinates $\tau_i: E_i \rightarrow \Lambda_i \subset \mathbb{R}$ may be used as representations. Their numerical values are not taken as primitive observables. Membership of Λ_i in $\mathbb{R}$ carries no assumption of density, connectedness, completeness, or continuum structure. Strictly increasing reparametrizations are allowed when only order matters; positive affine gauges are used only when affine calibration is part of the model.

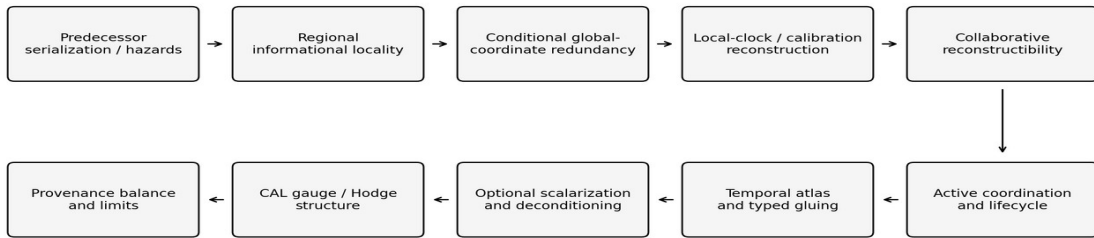
Locality constraints are structural rather than geometric: support, ports, and fanout are bounded independently of system size, spectator systems must not alter a local rule, and the construction must be equivariant under relabeling. No spatial coordinates, embedding, metric, dimension, or preferred geometry is introduced at this stage.

The three principal relation types are kept noninterchangeable:

- ORD: directed event precedence. Only ORD contributes to the collective partial order.
- CAL: orientation-preserving affine calibration between local clock representations. CAL compares representations but does not by itself create event precedence.
- CORR: a declared correlation witness. CORR alone creates neither precedence nor affine calibration.

The resulting hierarchy is conceptual rather than a claim of inevitable dynamics: Figure 1 summarizes the resulting successor architecture.

regional information $\rightarrow$ local-clock reconstruction $\rightarrow$ collaborative reconstructibility $\rightarrow$ active coordination $\rightarrow$
temporal atlas $\rightarrow$ global scalarization only when integrability permits it. (1)



Representation is separated from reconstruction; reconstruction is separated from collaborative irreducibility; temporal domains enter only after these earlier layers.

FIG. 1. Logical architecture of the finite successor theory. Regional informational locality precedes the conditional representation result; local-clock reconstruction then precedes the stronger collaborative layer. Temporal domains enter only after these earlier stages. The arrows denote logical dependency, not ontological reduction.

III. COLLABORATIVE RECONSTRUCTION AND GENERATION OF TEMPORAL INFORMATION

A. Reconstructive collaboration and collaborative complexity

Let Ω be a finite predeclared family of admissible histories and Q a target temporal relation. Participant i has only its declared local view V_i . For a set A of participants, write $V(A)$ for the joined information in those local views. We say that A reconstructs Q over Ω when Q is a function of $V(A)$ on that history family. The collaborative complexity is the smallest number of participant views with that property. This definition is epistemic/informational: it measures joint reconstructibility and is not identified with causal influence.

$$\kappa_{\Omega}(Q) = \min\{|A| : Q \text{ is determined by } V(A) \text{ on } \Omega\}. \quad (2)$$

Thus $\kappa=1$ means that at least one declared local view already suffices, whereas $\kappa \geq 2$ certifies a genuinely joint reconstruction requirement relative to Ω and the chosen view map. Larger κ means that a larger smallest set of views is jointly necessary; it does not by itself identify causal responsibility or individual variable essentiality. The cardinality minimization is used here as a finite determining-set functional; the claimed novelty lies in its role within the temporal construction, its explicit local generation, and its prospective robustness tests rather than in the abstract minimization problem alone.

B. Local microdynamics without a scalar clock or global target oracle

To separate collaborative reconstruction from an analyst who simply computes Q globally, the local constructive protocol uses finite token/ORD microdynamics whose updates use bounded local information, no primitive scalar time coordinate, and no global Q oracle. Within the constructive family tested, the dynamics generate instances with collaborative complexity $\kappa=2, \dots, 8$. This is a finite existence result: it shows that nontrivial distributed temporal information can arise in an explicitly local construction without being inserted as a global clock or target variable. It does not assert that the construction is unique, generic in all model classes, or a model of physical time in nature.

C. Exact product-state DAG result

For the exact identity below we now specialize to the declared DAG protocol, whose admissible participant-state family is a Cartesian product of the declared local state sets. This product-domain assumption is used only for Proposition 1 and is not part of the general definition of κ above.

For the declared finite DAG protocol with source s and sink t , let $Q_G \in \{\text{BEFORE}, \text{INCOMPARABLE}\}$: $Q_G = \text{BEFORE}$ iff the '+' participants contain a directed $s \rightarrow t$ path; otherwise $Q_G = \text{INCOMPARABLE}$. Participant i is Q -essential iff flipping only i changes Q_G for some admissible setting of the other states.

Proposition 1 (collaborative complexity on the declared product-state DAG protocol).

$$\kappa_{\Omega_G}(Q_G) = \# \text{Ess}(Q_G). \quad (3)$$

Proof. Every Q -essential participant must occur in any reconstructing set: its witness pair agrees on all retained views if that participant is omitted but has different target values. Conversely, let E be the set of Q -essential participants. If two admissible product states agree on E , product structure permits an admissible sequence between them that changes one nonessential coordinate at a time. By nonessentiality, Q is unchanged at each step, so the two endpoint states have the same target value. Hence E determines Q , and minimality follows from the witness argument.

The product-domain hypothesis is essential to the converse. On a constrained nonproduct history family, absence of single-coordinate flip witnesses need not imply that the target is determined by the individually essential coordinates. Equation (3) is therefore not exported beyond the declared product-state protocol without an additional connectivity/closure assumption.

The proposition separates collaboration from simple path counting. A shortcut can bypass otherwise relevant vertices and lower κ ; alternative redundant routes can make path members nonessential. The collaborative

content is therefore a property of the dependency logic induced over the declared history family, not merely of network size or path length.

D. External robustness holdout

The exact construction was followed by a one-shot, externally executed holdout on 288 predeclared synthetic cases. The substantive effect was $\kappa > 2$ rather than the minimal $\kappa = 2$ floor that can arise from source/sink structure. All predeclared criteria were satisfied, and an independent reconstruction reproduced the record-level outputs, summary, criterion evaluations, and final verdict. Table I summarizes the family-level outcomes. The claim used here is deliberately limited to finite-ensemble robustness.

TABLE I. One-shot external robustness holdout for collaborative reconstructibility. The controls show both positive and negative behavior: direct bypass collapses κ to 2, shortcut stress reduces κ in a subset of cases, and token loss can disconnect the network. The observed result is therefore sensitivity plus robustness, not saturation of a metric that always reports collaboration.

Family	Cases	Observed result
Positive chain control	16	16/16 with $\kappa = n$
Direct-bypass negative control	16	16/16 with $\kappa = 2$
Sparse baseline	64	64/64 with $\kappa > 2$
Shortcut stress	64	44/64 with $\kappa > 2$; 20/64 degraded to $\kappa = 2$
Token loss	64	43/64 remained connected; all 43 connected cases had $\kappa > 2$; 21 disconnected
Richer local routing rule	64	64/64 with nonconstant Q and 64/64 with $\kappa > 2$

This holdout supports a specific finite robustness claim. Within the predeclared ensembles and thresholds, $\kappa > 2$ persists across the unseen sparse baseline, a substantial fraction of shortcut-stress cases, every connected token-loss case, and all richer local-rule cases. The negative-control family with a direct bypass gives $\kappa = 2$ as designed. This does not establish that all temporal information is collaborative, that κ measures causal responsibility, or that physical time is generated by the tested protocol. It establishes that the collaborative phenomenon defined above is nontrivial, structurally responsive, and reproducible within the declared finite domain.

Proposition 1, the explicit local construction, and the external holdout play different evidential roles and are not collapsed into one claim. The proposition is exact for the declared product-state DAG protocol; the token/ORD microdynamics provide an explicit finite construction; and the holdout tests robustness over predeclared unseen ensembles. Together they motivate the term collaborative relational time used in the title: a temporal relation can be globally available to the joint system while remaining unavailable to every insufficient subset of local views, without requiring a primitive scalar clock. The physical significance of that structure remains an open question rather than an assumption.

IV. DYNAMIC COORDINATION, HISTORICAL MEMORY, AND ACTIVE COORDINATION DOMAINS

A. Two temporal layers: historical order and active coordination

Collaborative reconstructibility does not by itself define a temporal domain. To describe changing coordination, the model carries two different relational layers. The historical graph P_H records certified ORD and provenance that have already occurred; its role is archival and order-theoretic. The active graph A_H records relations that are presently available for coordination. An active temporal domain is then an analyst-

level connected component of A_H satisfying the declared internal coherence conditions. It is neither identical to P_H nor assumed to be maximal, spatially localized, or ontologically fundamental.

The distinction matters dynamically. **ACTIVATE** and **DEACTIVATE** change current coordination, whereas **LOCAL_COMMIT** and **TRANSFER_COMMIT** add new locally certified event relations or transfers. If an active bridge is removed, past cross-domain ORD and provenance may remain in P_H , but no new post-split comparison is generated merely because an earlier interaction once existed. A later re-merger must be supported by a fresh active relation or forward transfer. In this precise sense, historical memory is retained without being promoted to future influence.

B. Non-resurrection and finite lifecycle validation

The structural rule above is deliberately stronger than keeping a historical log: a split changes which future relations can be generated even though the past remains available for audit. A prospectively corrected finite lifecycle holdout satisfied all predeclared criteria within its declared domain. The study supports the split/remerge and non-resurrection semantics of the implementation; it is not evidence that physical systems must organize time in this way. The preceding failed lifecycle attempt is retained in Appendix B rather than rewritten retrospectively.

The conceptual consequence is limited but important: a temporal domain is a state of current coordination, not a permanent equivalence class generated by all past interactions. This prevents the historical record from functioning as a hidden universal synchronization channel.

V. TEMPORAL ATLASES: EXACT GLUING CRITERIA AND DYNAMIC ROBUSTNESS

A. Exact ORD and affine-CAL gluing criteria

Active domains may be treated as local temporal charts D_α carrying certified ORD relations and, when calibration is meaningful, positive-affine CAL transition maps. Global coherence is type-dependent. Pairwise compatibility alone is not sufficient: ORD must be checked for global acyclicity, whereas redundant affine CAL data must satisfy cycle-composition constraints.

Proposition 2 (finite temporal-atlas gluing criterion).

- **ORD:** a family of charts glues to a collective partial order if and only if the union of all certified ORD edges is acyclic. When this holds, the collective order is the transitive closure of that union.
- **CAL:** on a connected overlap graph, a compatible global affine scalarization exists, unique up to one global affine gauge, if and only if every affine cycle holonomy is the identity.
- Every connected CAL tree is scalarizable because a root gauge can be propagated recursively and there is no redundant cycle constraint.
- Nonidentity affine holonomy obstructs one global affine clock even when the ORD sector remains a coherent partial order.
- Whether a cycle holonomy is the identity is invariant under independent positive-affine reparametrizations of the local charts.

Proof sketch. For ORD, any directed cycle in the union would violate antisymmetry of a putative collective partial order; conversely, if the finite union is acyclic, its transitive closure is a partial order extending every local certified relation. For CAL, choose a root chart and propagate affine gauges along a spanning tree. Any non-tree edge then closes a cycle. The propagated assignment is independent of path exactly when every such cycle composition is the identity; equivalently, every cycle holonomy is trivial. A local affine reparametrization conjugates the composed transition around a cycle, so identity versus nonidentity is gauge invariant.

Thus relational temporal coherence and global scalarizability are logically distinct. A coherent ORD atlas may coexist with CAL holonomy, and a curl-free local calibration sector may still contain global obstruction, as the Hodge formulation below will make explicit.

An acyclic ORD atlas always admits order-preserving topological serializations, but those labels are not the affine CAL scalarization discussed here. The holonomy obstruction concerns compatibility of calibrated local clock representations, not the existence of some arbitrary numbering of a finite partial order.

B. Dynamic/noisy atlas validation and preserved failure

The exact proposition does not by itself establish robustness under dynamic or noisy chart data. That separate implementation question was tested in a fresh prospectively repaired holdout, which satisfied all predeclared criteria within the declared finite domain. The result supports the atlas/gluing distinctions under the tested perturbations; it does not enlarge Proposition 2 into a universal theorem. The earlier failed noisy-atlas attempt is preserved in Appendix B.

VI. SEMANTIC FIREWALLS, FINITE SCALE COMPATIBILITY, AND SCALAR DECONDITIONING

A. Semantic firewalls fixed before macroscopic interpretation

Before interpreting atlas behavior across scale, a dedicated semantic consistency check fixed a set of non-identifications that function as firewalls for the successor theory:

- reconstructibility $\neq$ causal influence;
- incomparability $\neq$ operational independence;
- active connectivity $\neq$ historical precedence;
- active connectivity $\neq$ directed reachability;
- memory $\neq$ future influence;
- CAL holonomy $\neq$ ORD cycle;
- affine holonomy $\neq$ spatial or gravitational curvature;
- affine CAL scalarizability is an optional reduction, not the definition of temporal organization.

These are constraints on inference, not stylistic caveats. In particular, a large reconstructibility complexity κ cannot be read as causal responsibility; an active connected component cannot be substituted for a historical causal order; and a failure of affine scalarization cannot be relabeled as spacetime curvature. The later scale and field results are interpreted only inside these firewalls.

B. Finite-scale compatibility of a macroscopic affine coordinate

A finite-scale validation study asked a positive question: can a macroscopic affine coordinate remain compatible with the declared local relational structure over a predeclared range of finite sizes? The study satisfied its predeclared criteria. This licenses only finite-domain compatibility. It neither proves that such a coordinate remains uniformly well conditioned as the system grows nor supplies an $N \rightarrow \infty$ or continuum limit.

C. Relational coherence despite global scalar deconditioning

A complementary scalar-deconditioning study then asked a different question: can bounded-radius relational coherence remain stable while one global scalar representation becomes progressively more weakly conditioned? In the tested chain-plus-span-2 calibration family, local mismatch predictions remained essentially unchanged while endpoint effective resistance R_{eff} , used as a scalar-synchronization susceptibility, increased strongly with system size. Table II reports the finite-size values, and Fig. 2 visualizes the separation.

Here R_{eff} is the endpoint effective resistance of the weighted calibration graph and is used only as a finite conditioning diagnostic for inference from relative measurements, consistent with the electrical-network interpretation of relative-estimation error (Ref. 15). The term “susceptibility” is descriptive; no thermodynamic susceptibility is implied.

TABLE II. Finite scalar-deconditioning example. The log-log slope of R_{eff} versus N is 0.986606 over the tested sizes. All 48 unseen local-mismatch cases satisfied the predeclared relational-coherence criterion. No asymptotic or continuum inference is made.

N	Local prediction p90	Endpoint scalar susceptibility R_{eff}
24	0.0099491	4.9577709
48	—	9.7577709
96	—	19.3577709
192	0.0098429	38.5577709

The finite-scale compatibility and scalar-deconditioning studies are complementary rather than contradictory. The first shows that a compatible macroscopic affine reduction can exist over a declared finite scale range; the second shows that existence or local compatibility does not imply uniformly good global conditioning. Within this tested family, the declared local relational coherence is therefore more stable than the numerical conditioning of the chosen global affine reduction; no stronger constitutive or asymptotic claim is made.

No asymptotic claim follows from the observed near-linear growth of R_{eff} . The result is a finite separation-of-properties statement: over the tested sizes and predeclared controls, local coherence and global-scalar conditioning do not track one another.

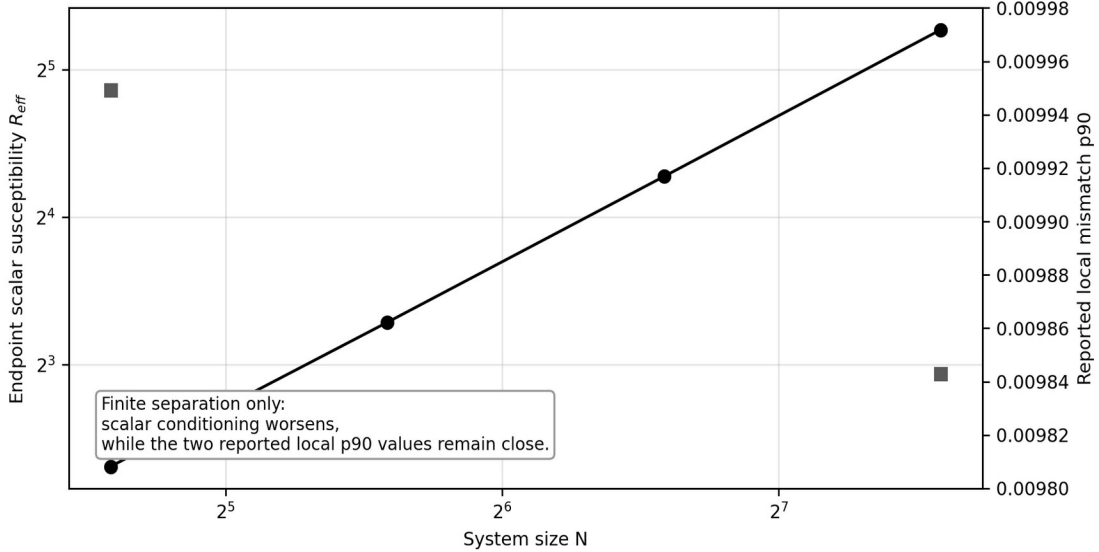


FIG. 2. Finite scalar deconditioning. Endpoint effective resistance increases over the tested sizes. The local-mismatch series contains only the two actually reported p90 values, at $N = 24$ and $N = 192$; no intermediate values are interpolated or inferred.

VII. TYPED INTERACTION: PARTIAL GLUING WITHOUT GENERIC SYNCHRONIZATION

A. Exact consequences of ORD, CAL, and CORR typing

Only after collaborative reconstruction, active coordination, and atlas gluing have been defined do temporal domains become useful as an interaction layer. Two internally coherent active patches need not share the same temporal relation merely because they are connected. The effect of an inter-domain link depends on its declared type and consistency:

- ORD: none, partial precedence, or an inadmissible cycle.
- CAL: unidentified relative gauge, single-link nonredundant identification, redundantly scalarizable calibration, or holonomy obstruction.
- CORR: witness only; no precedence or calibration is generated by CORR alone.

Proposition 3 (typed partial gluing).

- Disconnected scalarizable domains retain independent affine gauge freedoms.
- Cross-domain ORD creates partial comparison without identifying CAL gauges.
- One CAL bridge identifies relative affine gauge but is nonredundant.
- Redundant CAL constraints are jointly scalarizable exactly when inferred calibrations are cycle-consistent.
- A connected CAL subgraph that is a tree is globally affine-scalarizable; redundant CAL cycles provide falsifiable holonomy constraints.
- Removing active inter-domain relations splits active coordination without resurrecting future cross-domain ordering.
- Fresh active relation(s) of the relevant coordination type are required for reglue; CORR alone does not create ORD or CAL closure.
- CORR-only coupling creates neither precedence nor affine calibration.

Proof sketch. ORD and CAL participate in different closure operations. Adding an ORD bridge enlarges directed reachability and may therefore create only those cross-domain comparisons implied by transitive closure; it does not identify affine chart gauges. A single CAL bridge fixes a relative affine map but supplies no redundant consistency test. Additional CAL links close cycles, so joint scalarizability is equivalent to consistency of the induced cycle compositions. Removing the active bridges deletes the present coordination relation but does not delete historical records, hence future ordering does not resurrect without a new active event. CORR is excluded by definition from both ORD closure and CAL composition, so CORR-only connectivity has neither effect.

B. Matched-connectivity specificity test

A direct confound remains: perhaps any connected inter-domain skeleton would be classified as sharing a common temporal structure, making relation typing decorative. A matched-connectivity holdout attacked this possibility by holding connectivity fixed and varying only the relation semantics. In the 200-case unseen holdout over $D=3,4,5,6$ domains, the ORD multiple-link, CAL-tree, and CORR-only families used the same chain $0-1-\dots-(D-1)$ with the same number of links. Table III displays the matched-connectivity comparison.

TABLE III. Matched-connectivity semantic-specificity control. With the skeleton fixed, the implementation preserves the distinct closure semantics assigned to ORD, CAL, and CORR.

Relation on identical chain	Observed temporal semantics
ORD multiple-link chain	Partial cross-domain precedence; no affine calibration
CAL tree	Connected common relative affine gauge; scalarizable but nonredundant
CORR-only chain	No cross-domain precedence; relative affine gauge unidentified

The corrected external attempt satisfied all predeclared criteria. Consistent redundant CAL cycles were scalarizable, an inconsistent redundant CAL link produced a holonomy obstruction, split/non-resurrection and fresh-reglue cases satisfied their checks, and CORR-only null cases remained temporally null. This is an implementation and semantic-specificity validation: because the relation types are defined to obey different closure rules, the test does not discover a new physical law. It instead rejects the concrete confound that the tested classifier or dynamics might effectively be using connectivity alone. The pre-science contract defect of the earlier attempt is retained in Appendix B.

At this point the successor chain has established the ingredients needed for a field-like calibration description: local affine gauges exist where CAL is meaningful, global scalarization is optional, and nontrivial cycle inconsistency is a gauge-invariant obstruction. The next sections therefore study the positive CAL scale sector as a finite cochain field, without introducing spatial geometry.

VIII. CAL SCALE CONNECTION AND LOCAL CLOCK GAUGE

A. Positive CAL scale sector

A full CAL map is affine and contains both scale and translation. The field construction isolates the positive scale sector because multiplication becomes additive under the logarithm. Translation holonomy remains part of full affine gluing but is not included in the linear Hodge decomposition developed here. For an oriented CAL edge $i \rightarrow j$ define

$$\gamma_{ij} = \log a_{j \leftarrow i}. \quad (4)$$

Under independent local positive clock rescalings $\tau'_i = c_i \tau_i$, with $q_i = \log c_i$, the relative scale transforms by $a'_{j \leftarrow i} = c_j a_{j \leftarrow i} c_i^{-1}$; therefore

$$\gamma' = \gamma + D_0 q. \quad (5)$$

Equation (5) is the local scale-gauge law. Any candidate observable in this sector must either be invariant under addition of $D_0 q$ or be explicitly recognized as gauge-dependent.

B. Registered compositional 2-cells and curl

A graph alone has no canonical local curl. The atlas must register finite compositional CAL 2-cells whose boundary composition has relational meaning. These are calibration-consistency cells, not spatial faces. Let D_1 be the edge-to-cell coboundary. The boundary-of-boundary identity is

$$D_1 D_0 = 0. \quad (6)$$

The registered-cell temporal curl is

$$F = \text{curl}_T \gamma := D_1 \gamma. \quad (7)$$

By Eq. (6), F is gauge-invariant. On a bare graph without registered 2-cells one may still discuss integrated cycle periods or affine holonomy, but one may not infer a local face-curl observable merely from graph topology.

IX. EXACT, CIRCULATORY, AND HARMONIC TEMPORAL SECTORS

On a finite registered relational 2-complex with the unit-weight cochain inner product, γ admits a discrete Hodge decomposition; Refs. 4–7, 10, and 11 provide the relevant finite-complex and Hodge background. The inner product selects orthogonal representatives in cochain space and is not interpreted as a spatial metric. Related connection and synchronization constructions provide a useful mathematical comparison for the CAL sector (Refs. 8 and 9), although the present observables and claims are different.

$$\gamma = D_0 s + D_1^T \psi + h, \quad (8)$$

$$h \in \ker(D_0^T) \cap \ker(D_1). \quad (9)$$

The exact component $D_0 s$ is integrable into a scalar potential. The coexact component $D_1^T \psi$ carries registered local circulation. The harmonic component is closed and coclosed, can be invisible to local curl, and can retain a nonzero global period.

The exactness obstruction and the underlying cohomology class are independent of the choice of harmonic representative. By contrast, the orthogonal decomposition into particular representatives, and all norms introduced below, depend on the declared cochain inner product. The unit-weight choice is therefore part of the finite model specification, not a derived physical metric.

Theorem 4 (scale scalarization criterion).

On a connected fixed CAL complex, a common global log-scale potential exists exactly when $\gamma \in \text{im}(D_0)$. Equivalently, the coexact and harmonic components of Eq. (8) both vanish.

A key consequence is that curl-free does not imply globally scalarizable. When H^1 is nontrivial, $D_1\gamma=0$ can coexist with a nonzero harmonic class and nonzero cycle period. Local integrability and global integrability are therefore distinct.

X. STATIC DIVERGENCE NO-GO AND GAUGE-INVARIANT FIELD CONTENT

A. Why the naive divergence fails

The most tempting source observable is $-D_0^T\gamma$. It fails the gauge test. More generally, for a positive diagonal edge-weight matrix W and $A=-D_0^TW$,

$$A(\gamma + D_0 q) = A\gamma - D_0^T W D_0 q. \quad (10)$$

Theorem 5 (static divergence no-go).

On a connected nontrivial CAL graph with positive edge weights, no nonzero weighted incidence divergence $-D_0^TW\gamma$ is gauge-invariant under independent local scale gauges. Gauge invariance would require $D_0^TW D_0 q=0$ for every q , but the positive weighted graph Laplacian is strictly positive on nonconstant gauge modes.

Registered curl does not rescue a nonzero static vertex source: $J_{circ}=D_1^T F$ is gauge-invariant but $D_0^T J_{circ}=(D_1 D_0)^T F=0$. Static CAL circulation therefore cannot by itself furnish a nonzero vertex source density.

B. Transverse/nonintegrable content

Let $P_\perp$ be the orthogonal projector onto the complement of $\text{im}(D_0)$, and define

$$\eta = P_\perp \gamma, P_\perp D_0 = 0. \quad (11)$$

Then η is invariant under Eq. (5) and vanishes exactly when the scale connection is globally scalarizable on the connected fixed complex. This statement is made for the declared cochain inner product: changing that auxiliary inner product can change the orthogonal representative and its norm, although vanishing of the projection continues to characterize membership in $\text{im}(D_0)$. We use η as an analyst-level diagnostic; the microdynamics is not assumed to compute a global projector.

$$E_{inv} = \|\eta\|^2, \rho_i = \frac{1}{2} \sum_{e \ni i} \eta_e^2. \quad (12)$$

The local shares ρ_i are gauge-invariant and sum to E_{inv} . On a fixed complex the invariant content decomposes further into coexact and harmonic contributions. Across an update, $\sigma_i = \rho_i^+ - \rho_i^-$ and $\Delta E_{inv} = \sum_i \sigma_i$ remain gauge-invariant even when the endpoint clock gauges are chosen independently.

The half-incidence assignment used for ρ is a symmetric bookkeeping convention rather than a unique local observable. Other gauge-invariant partitions of the same edge content are possible. Accordingly, local transport statements below are conditional on both the declared cochain inner product and the declared density partition; the total quadratic content is the less convention-dependent quantity once the inner product is fixed.

XI. PROVENANCE-INDUCED CURRENT AND CONDITIONAL BALANCE

A. From density change to a current assignment

Gauge-invariant density change alone does not distinguish production from redistribution. A continuity equation admits nonunique currents on a cyclic graph because divergence-free cycle currents may be added. To define one particular current assignment, we therefore condition on the micro-event provenance supplied

by the declared model. This is a routing convention anchored in provenance, not a new clock gauge and not an independently derived microscopic transport law.

For one micro-event e , let T_e be a finite rooted provenance tree with write-anchor/root r_e . For each oriented provenance edge $p \rightarrow c$ define

$$J_e(p \rightarrow c) = \sum_{v \in \text{subtree}(c)} \Delta \rho_v. \quad (13)$$

With signed tree divergence (outgoing minus incoming),

$$\Delta \rho + \text{div}_T J = S. \quad (14)$$

Theorem 6 (provenance-tree balance theorem).

Conditional on a finite rooted event-provenance tree and write-anchor supplied by the micro-event law, Eq. (13) yields the unique current satisfying Eq. (14) under the root-source convention. Every nonroot vertex has $S_v = 0$, while the root carries $S_r = \sum_v \Delta \rho_v$. On a cyclic provenance graph, uniqueness fails up to divergence-free cycle currents.

The resulting current assignment is gauge-invariant because it is reconstructed from gauge-invariant $\Delta \rho$ and supplied provenance. It is relabel covariant and spectator independent under the declared routing rule. Changing the provenance/root can change localization of the assigned current but not the total residual. The microdynamics is not assumed to compute this observer-level reconstruction.

B. Integrated balance and the open Noether question

Define the total invariant content

$$Q_T := \sum_i \rho_i = E_{inv} = \|\eta\|^2. \quad (15)$$

Summing Eq. (14) over a closed provenance patch cancels all internal currents and gives

$$\Delta Q_T = \sum_i S_i. \quad (16)$$

Corollary 7 (source-free algebraic balance).

If the closed patch is source-free, $\sum_i S_i = 0$, then $\Delta Q_T = 0$ exactly. This is an algebraic consequence of the finite continuity construction. No action principle has been assumed, so Q_T is not identified as a Noether charge. A variational extension would require separate analysis of local gauge identities (Noether II) and of any genuine continuous global symmetry capable of producing a Noether-I charge (Ref. 12).

XII. VALIDATION SCOPE AND REPRODUCIBILITY

The mathematical statements and computational studies in this article have different evidential status. Analytic distinctions delimit what a claim means; propositions and theorems are exact within stated finite assumptions; constructive validation tests declared mechanisms; predeclared synthetic holdouts test implementation robustness over declared ensembles; and reproducibility records preserve nonretroactivity. None of these layers is treated as experimental evidence that nature realizes the model.

Table IV gives the public claim-to-evidence map at manuscript level. Detailed versioned artifact identifiers, hash manifests, record-level outputs, independent reconstruction records, and negative-path history are retained in the reproducibility archive. Their function is traceability and reproducibility, not a cumulative “proof score.”

TABLE IV. Manuscript-level claim-to-evidence map. Counts describe finite declared validation domains; they do not convert the corresponding claim into a universal or experimental statement.

Scientific layer	Evidence type	Finite support and claim boundary
Regional informational locality	Analytic distinction	Separates support, regional sufficiency, arbitration access, and trajectory-dependent causal domain; no common-origin reachability claim.
Conditional global-coordinate redundancy	Finite validation	32/32 cases (16 training + 16 holdout); 16/16 criteria; replay. Collective temporal structure is supplied, not derived.
Local-clock/calibration reconstruction	Constructive validation	Validated suites include 60/60 exact factorized, 80/80 dynamic-interface, 112/112 correlated-domain, and 240/240 approximate-closure cases. Mixed-record/rollback layers remain finite and interface-dependent.
Cross-interface composition	Finite validation	156/156 cases; 156/156 agreement between evaluation paths; 48/48 multipath concordance checks. Implementation consistency only.
Collaborative reconstructibility	Exact proposition + external holdout	Exact product-state DAG result plus 288-case synthetic holdout with positive and negative controls; finite robustness, not causal responsibility.
Active lifecycle	Structural rules + external holdout	320 unseen cases; historical integrity 280/280 noncontrol cases and 3688/3688 qualifying isolation checks; finite implementation semantics.
Atlas, scalar conditioning, and typed interaction	Exact propositions + finite validation	Fresh prospective dynamic/noisy validation; 48 unseen local-mismatch cases in the deconditioning test; 200 matched-connectivity cases for ORD/CAL/CORR specificity.
CAL gauge/Hodge and provenance balance	Exact finite algebra	Gauge law, Hodge decomposition, divergence no-go, and provenance-tree balance. Depends on declared inner product/density/provenance conventions; no Noether claim.

XIII. INTERPRETATION, SCOPE, AND FALSIFIABLE EXTENSIONS

The finite results support a layered view of temporal organization. Local ORD supplies precedence. Overlapping dependencies can make target temporal information collaboratively reconstructible. Active coordination reorganizes without erasing history, yet history alone does not force future interaction. Compatible CAL can integrate to a common affine representation, but that scalarization is a reduction rather than a primitive. Nontrivial holonomy or harmonic periods can obstruct the CAL reduction while leaving other temporal relations coherent. Gauge-invariant nonintegrable calibration content can then be assigned a declared local density and, conditional on provenance, a current assignment.

Several boundaries are essential. First, all results are finite; no continuum or $N \rightarrow \infty$ limit is established. Second, regional informational sufficiency is not a trajectory-dependent causal-domain theorem, and the regional witness construction does not establish common-origin dynamical reachability. Third, conditional redundancy of a global scalar coordinate assumes collective ordering/calibration structure; the staged local-clock program addresses reconstruction of that structure only within its declared finite interfaces. Fourth, the relational complexes carry no primitive spatial geometry: no coordinates, distances, dimension, embedding, volume, Lorentzian metric, causal cone, curvature tensor, stress-energy tensor, or Einstein equation is derived. Registered CAL 2-cells are compositional consistency cells, not spatial faces. Although finite ORD partial orders are mathematically adjacent to causal-set constructions (Ref. 14), no causal-set dynamics, manifold approximation, or spacetime interpretation is claimed.

Reconstructibility is not causal influence. The collaborative complexity κ quantifies how many declared views are jointly necessary to reconstruct Q over the stated history family; it does not say that each participant dynamically caused Q . Active coordination domains are analyst-defined patches, not ontological “islands of time.” Affine CAL holonomy is a calibration obstruction and is not identified with gravitational or spatial curvature. The transverse representative, its quadratic norm, and the vertex density depend on declared finite conventions (cochain inner product and density partition) in addition to being gauge-invariant. Finally, the provenance-induced current is a conditional reconstruction; its source residual is not creation or destruction of physical time.

The most immediate falsifiable extensions are therefore not broader metaphysical claims but sharper structural tests: topology-changing transport across changing complexes; noisy or incomplete registration of CAL cells; provenance selection on cyclic DAGs; and a variational/symmetry analysis that can either derive or rule out a Noether interpretation of Q_T . A no-go result on any of these fronts would delimit the framework without retroactively invalidating the exact finite statements proved here.

A particularly important next test is convention robustness: repeat the transverse-content and continuity analysis under a predeclared family of positive edge inner products and admissible gauge-invariant density partitions. Persistence of qualitative conclusions would strengthen the field-like interpretation; strong dependence would instead identify the present unit-weight/half-incidence construction as a useful but nonunique coordinate on invariant content.

XIV. CONCLUSION

The successor program establishes a coherent finite chain rather than a single emergence claim. Regional analysis first separates informational sufficiency from trajectory-dependent causal domain. Conditional on supplied collective temporal relations, a global scalar coordinate and total serialization are representationally redundant in the declared finite validation. A staged local-clock/calibration program then reconstructs collective chronology across exact, dynamic, correlated-domain, mixed-record, and approximate-closure layers. Only after that reconstruction problem is separated do bounded local dependencies generate examples of genuinely joint target information, quantified by collaborative complexity and tested by a predeclared 288-case external synthetic holdout.

Dynamic coordination then supplies lifecycle semantics; temporal atlases determine when local structures glue; finite-scale tests separate local relational coherence from conditioning of a chosen affine scalar reduction; and matched-connectivity controls verify that the implementation respects declared relation type rather than bare connectivity. These are finite structural and implementation results, not evidence that nature is obliged to instantiate the ORD/CAL/CORR catalogue.

The CAL scale sector subsequently admits a gauge-covariant cochain γ , registered-cell curl, and exact/coexact/harmonic decomposition. The static divergence no-go prevents gauge-dependent clock choices from being misread as temporal sources. Projection onto the transverse sector yields gauge-invariant nonintegrable content η and density ρ , while supplied provenance defines an observer-level current assignment J satisfying $\Delta\rho + di v_T J = S$. The integrated source-free balance conserves Q_T algebraically, with the Noether question left open.

The current and density used in this final layer remain conditional on the declared inner product, local density partition, and provenance convention. Their value is therefore as a controlled gauge-invariant finite construction, not yet as a unique physical field theory.

Within the declared finite framework, the operative structure is therefore not a universal scalar time value but a typed web of local event relations and dependencies from which collective temporal organization can be reconstructed, generated, reorganized, and, in the CAL sector, sometimes reduced to a common affine representation. An ORD partial order may still admit arbitrary order-preserving serializations; a compatible global affine CAL clock exists only when the calibration structure satisfies the corresponding integrability conditions.

AUTHOR DECLARATIONS

Conflict of Interest

The author has no conflicts to disclose.

Ethics Approval

Not applicable; this study involves no human participants, animals, or identifiable personal data.

Author Contributions

Mariano Boj Viudes: Conceptualization (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Software (lead); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The reproducibility materials corresponding to this manuscript are archived on Zenodo under version DOI <https://doi.org/10.5281/zenodo.21938420>, within concept record <https://doi.org/10.5281/zenodo.21931499>. Earlier versioned records are <https://doi.org/10.5281/zenodo.21931500> and <https://doi.org/10.5281/zenodo.21932410>. The v2.0.0 archive includes machine-readable SHA-256 manifests, recovered primary successor evidence, checkpoint-bound historical records, validation outputs, and claim-evidence/provenance documentation. Historical records remain frozen and are not retroactively modified.

APPENDIX A: PROOF DETAILS FOR GAUGE AND BALANCE IDENTITIES

1. Gauge law for the scale sector

Let $\tau'_i = c_i \tau_i (c_i > 0)$. If $\tau_j = a_{j \leftarrow i} \tau_i + b_{j \leftarrow i}$, then $a'_{j \leftarrow i} = c_j a_{j \leftarrow i} c_i^{-1}$. With $q_i = \log c_i$ and $\gamma_{ij} = \log a_{j \leftarrow i}$, taking logarithms gives $\gamma' = \gamma + D_0 q$.

2. Hodge decomposition and harmonic obstruction

For a finite cochain complex $C^0 \xrightarrow{D_0} C^1 \xrightarrow{D_1} C^2$ with $D_1 D_0 = 0$ and the unit-weight cochain inner product, $C^1 = \text{im}(D_0) \oplus \text{im}(D_1^T) \oplus [\ker(D_0^T) \cap \ker(D_1)]$. The last factor represents H^1 by its harmonic representative; hence a closed field can fail to be exact precisely when its harmonic class is nonzero.

3. Positive-weight divergence no-go

For $A = -D_0^T W$ with W positive diagonal, gauge invariance would require $A D_0 = 0$. But for any nonconstant q on a connected graph, $q^T D_0^T W D_0 q = \|W^{1/2} D_0 q\|^2 > 0$. Therefore $A D_0$ cannot vanish on all gauge modes.

4. Provenance-tree continuity

For each nonroot vertex v with parent $p(v)$, define $J_{p(v) \rightarrow v} = \sum_{u \in \text{subtree}(v)} \Delta \rho_u$. The incoming current at v is the subtree sum and the outgoing currents are the child-subtree sums. Since $\text{subtree}(v)$ is the disjoint union of $\{v\}$ and its child subtrees, outgoing minus incoming equals $-\Delta \rho_v$, so $S_v = \Delta \rho_v + d(v) J_v = 0$. At the root the residual is $\sum_v \Delta \rho_v$. The reduced incidence matrix of a tree has full rank, giving uniqueness under the root-source convention.

5. Integrated balance and the open Noether question

Summing $\Delta \rho + d i v_T J = S$ over all vertices of a closed provenance patch cancels every internal current once with each sign. Hence $\Delta Q_T = \sum_i S_i$ with $Q_T = \sum_i \rho_i = \|\eta\|^2$. If $\sum_i S_i = 0$ then Q_T is conserved exactly. This is an algebraic corollary of the provenance continuity law and not a Noether derivation.

APPENDIX B: REPRODUCIBILITY AND NEGATIVE-PATH RECORD

As in the predecessor article, machine-oriented provenance is kept outside the scientific narrative. The reproducibility materials for this manuscript record versioned source packages, predeclared case sets and criteria, record-level outputs, independent reconstructions, replay results where applicable, SHA-256 manifests, and the negative-path history used to delimit the accepted evidence.

Several negative paths are retained without retrospective repair. An initial interface-composition attempt was invalidated before it could support a scientific claim because generator/runtime-oracle, tolerance, holdout, multiplicity, metric, and oracle-conflict defects were found. A first lifecycle holdout genuinely failed before the prospectively corrected study. A first noisy-atlas holdout likewise failed before a new repaired prospective design was executed. A first matched-connectivity attempt was abandoned before scientific execution after a preflight contract defect was found.

No failed or invalidated attempt is reinterpreted as supporting evidence. Detailed internal identifiers and cryptographic hashes are therefore omitted from the manuscript text and maintained instead in machine-readable archive metadata, where they serve provenance and exact-replay functions rather than scientific notation.

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